From-Scratch Small Language Models for Legal and Financial Text: A Staged Pretraining Recipe and an Honest Account of Their Limits

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Abstract

We build two decoder-only language models from scratch — 125.8M and 528.5M parameters — pretrained on US case law, SEC filings, and educational web text, then fine-tuned for legal and financial question answering. We report three findings we believe are broadly useful beyond this specific project. First, naively mixing five heterogeneous pretraining sources at their natural size proportions measurably *hurts* held-out quality on the original, smaller sources (fineweb-edu perplexity worsens from 20.73 to 22.82 despite more total training data); a staged Warmup-Stable-Decay schedule, adapted from MiniCPM and SmolLM2, recovers this and improves perplexity on every shared domain over the smaller model. Second, a subtle supervised fine-tuning failure mode — two datasets sharing clause-category names under different task framings (extraction vs. classification) — causes the model to collapse to bare label output regardless of what is asked; we document the bug, its 100% reproduction rate, and the fix. Third, and most centrally, we run both models through five externally accepted benchmark categories via `lm-evaluation-harness` [Gao et al., 2024] and find a sharp, consistent split: open-book contract-clause tasks (given a passage, answer about it) are a real, usable capability, while closed-book legal knowledge and reasoning (MMLU legal subsets, CaseHOLD) sit at or *below* random chance for both model sizes. We further report a methodological case study on legal-domain contamination checking: a naive n-gram overlap check flags an implausible 8,712 lines as "contaminated," which on inspection are uniformly generic legal boilerplate (standard-of-review language, ToS disclaimer text, canonical precedent quotations) rather than evidence of benchmark leakage; a stricter 40-word contiguous-run threshold combined with per-hit provenance attribution narrows this to 5 of 1,853 documents (0.27%), each independently explainable and none affecting the reported results. We release both models, the fine-tuning data recipe, an in-browser ONNX deployment, and all evaluation code.

1 Introduction

Small (sub-billion-parameter) language models trained from scratch on a narrow domain are cheap to build and easy to deploy, including entirely client-side in a browser, but their actual capabilities and failure modes are not always reported with the same rigor as frontier-model releases. This paper documents an end-to-end build of two such models for legal and financial text – a 125.8M-parameter model and a 528.5M-parameter successor – with a specific focus on three questions that matter more for a deployed small model than raw benchmark leaderboard position: (1) does adding more pretraining sources actually help, or can naive mixing hurt it; (2) what training failure modes appear at this scale that would not be caught by aggregate loss curves alone; and (3) once a model is fine-tuned into a question-answering assistant, does it hold up against external, accepted evaluation, not just an in-house test set built by the same team that built the model.

Our contributions:

1. Two from-scratch Llama-style decoder language models [Touvron et al., 2023] at 125.8M and 528.5M parameters, with a shared byte-level BPE tokenizer, trained on US case law, SEC filings, and educational web text.
2. A staged (Warmup-Stable-Decay) pretraining data schedule that fixes a directly measured regression from naive proportional mixing across five heterogeneous sources, improving held-out perplexity on every domain shared between the two model generations.
3. A supervised fine-tuning recipe combining expert-annotated legal datasets (CUAD, LEDGAR) with teacher-distilled question answering data, including a documented failure mode – category-name collision between an extraction task and a classification task – and its fix.
4. A paired evaluation protocol (both model sizes answering the *identical* held-out questions) showing that scaling improves aggregate metrics without being uniformly better on every individual question.
5. External validation across five benchmark categories via `lm-evaluation-harness`, with an explicit methodological case study on why naive contamination checking over-flags legal text specifically, and how to fix it.
6. Public release of both models, the fine-tuning dataset recipe, an ONNX export for in-browser inference, and the full evaluation codebase, including the custom benchmark task configurations described in Section 7.

2 Related Work

Small and efficient language models. A line of recent work shows that careful data curation and training schedules can substitute for scale: Pythia, OLMo, TinyLlama, Phi, MiniCPM [Hu et al., 2024], and SmolLM2 [Allal et al., 2025] all report staged or curriculum-style data schedules as part of achieving strong small-model results. We adopt MiniCPM's Warmup-Stable-Decay (WSD) scheduling idea directly, applying it not to a learning-rate schedule but to per-source sampling weights, to fix a data-mixing problem specific to combining a small set of curated, high-value sources with much larger bulk sources.

Domain-specific legal and financial language models. BloombergGPT [Wu et al., 2023] and SaulLM [Colombo et al., 2024] both pretrain at much larger scale (50B and 7B+ parameters, respectively) on finance- and law-specific corpora. Our work asks a complementary question: how far can a domain-specific model go at two to three orders of magnitude fewer parameters, and where precisely does it stop working.

Legal NLP benchmarks. We evaluate against CUAD [Hendrycks et al., 2021b], an expert-annotated contract clause extraction dataset; LexGLUE [Chalkidis et al., 2021], which packages LEDGAR (clause classification) and CaseHOLD together with several other legal tasks; the original CaseHOLD dataset [Zheng et al., 2021], a 5-way multiple-choice legal holding identification task; and LegalBench [Guha et al., 2023], a 162-task collaboratively built benchmark for legal reasoning, from which we construct a 12-task contract-focused subset (Section 7.2). A concurrent study [Vaddi, 2026] evaluates sub-10B models (including a 3B-active mixture-of-experts model) on ContractNLI, CaseHOLD, and ECtHR under five prompting strategies including retrieval-augmented generation, finding that architecture and training quality matter more than raw parameter count within that range; our work looks two further orders of magnitude down in scale, at models an order of magnitude smaller than even that study's smallest model, without retrieval augmentation, to characterize what remains possible under those constraints.

Capacity and memorization. Allen-Zhu and Li [2024] estimate that language models store roughly 2 bits of knowledge per parameter, a hard ceiling largely independent of training recipe. Carlini et al. [2022] show memorization rate scales sharply with how many times a fact is repeated in training data, with facts seen once memorized at a small fraction of the rate of facts seen hundreds of times. Both results directly motivate our closed-book/open-book framing: a model this size cannot be expected to reliably recall specific, low-repetition legal facts from parametric memory, regardless of fine-tuning quality.

Retrieval-augmented generation. Lewis et al. [2020] introduce retrieval-augmented generation as a way to decouple a model's reasoning ability from its need to memorize facts. We view retrieval as a complementary, largely orthogonal direction to the one studied here: this paper characterizes purely parametric capability at small scale, which we believe is a necessary baseline before layering retrieval on top, and we leave a retrieval-augmented extension to future work.

General-purpose benchmarks and evaluation infrastructure. We use `lm-evaluation-harness` [Gao et al., 2024], the framework behind a widely used public LLM leaderboard, and its built-in HellaSwag [Zellers et al., 2019], ARC-Easy [Clark et al., 2018], PIQA [Bisk et al., 2020], and MMLU [Hendrycks et al., 2021a] tasks as a peer-comparison reference point against other small models, separate from our legal-domain-specific claims.

3 Architecture

Both models share a Llama-style decoder architecture [Touvron et al., 2023]: RoPE positional encoding [Su et al., 2021], SwiGLU activations [Shazeer, 2020], RMSNorm [Zhang and Sennrich, 2019], and tied input/output embeddings. The 500M model is a straightforward width-and-depth scale-up; both share the same 16,384-token byte-level BPE tokenizer, trained once and reused rather than retrained, so that tokenization is not a confound in the size comparison in Section 6.1.

4 Pretraining Data and the Staged Mixing Schedule

4.1 Sources

The 125M model pretrains on three sources: US case law (`HFforLegal/case-law`, CC BY 4.0), SEC 10-K filings (`PleIAs/SEC`), and educational web text (`HuggingFaceFW/fineweb-edu` [Penedo et al., 2024], ODC-BY), at a 35%/42%/23% mix, totalling 2.04B unique tokens over 2 epochs (4.08B tokens seen), reaching a held-out perplexity of 7.76.

The 500M model adds two large bulk sources – SEC 10-K/10-Q/8-K filings (`TeraflopAI/SEC-EDGAR`, Apache-2.0) and additional US case law (`common-pile/caselaw_access_project`, public domain) – for five sources totalling 14.7B unique tokens after cleaning, deduplication, and 13-gram decontamination against CaseHOLD and LexGLUE (see Section 8 for why this specific decontamination step matters). Given the larger unique-token budget, this model trains for a single epoch rather than two.

Table 1: Architecture comparison.

	125M model	**500M model**
Parameters	125,848,320	528,538,752
Layers	12	24
Hidden size	768	1,152
Attention heads	12	18
Head dimension	64	64
MLP (SwiGLU) intermediate size	3,072	4,608
Context length (tokens)	1,024	1,024
Vocabulary	16,384 byte-level BPE (shared)	
Positional encoding	RoPE, $\theta = 10{,}000$	
Normalization	RMSNorm, $\epsilon = 10^{-5}$	
Embeddings	tied input/output	
Precision	bf16 compute, fp32 master weights	

4.2 A measured regression from naive mixing

Our first attempt mixed all five sources proportionally to their natural size. Because the two new bulk sources are far larger than the three original sources, this diluted the original case-law/SEC/fineweb-edu "identity" of the smaller model down to roughly 13% of total training tokens combined. The result was not neutral: held-out perplexity on fineweb-edu, one of the three original sources, *worsened* from 20.73 (125M model) to 22.82 under this flat mix, despite the 500M model having more parameters and more total training data. Naive proportional mixing of heterogeneous sources is not a safe default; it can actively regress quality on the sources it dilutes.

4.3 Staged (Warmup-Stable-Decay) fix

Following the staged-schedule precedent in MiniCPM [Hu et al., 2024] and SmolLM2 [Allal et al., 2025], we replace the flat mix with a two-phase per-source sampling schedule: broad, size-balanced weights for the first 85% of training steps (*stable* phase), then a *decay* phase for the final 15% of steps that upweights the three original, curated sources specifically, so the model sees them last and retains them most strongly.

Table 2: Per-source sampling weights, stable vs. decay phase.

Source	**Stable weight**	**Decay weight**
fineweb-edu	0.15	0.30
case-law	0.13	0.25
SEC 10-K	0.15	0.25
SEC-EDGAR (bulk)	0.30	0.12
Case law (bulk)	0.27	0.08

This changes only which windows of already-tokenized data are sampled at each training step; it required no new data engineering, and did not change total step count, token budget, or wall-clock time relative to the flat-mix run.

4.4 Result: perplexity on every shared domain improves

Table 3: Held-out perplexity, 500M (staged schedule) vs. 125M. SEC-EDGAR and case law (bulk) have no 125M-model baseline, since those sources did not exist in the smaller model's pretraining mix.

Source	125M	500M (staged)
US case law	8.22	**7.23**
SEC 10-K	4.36	**3.96**
Educational web (fineweb-edu)	20.73	**17.81**
SEC-EDGAR (bulk)	—	3.84
US case law (bulk)	—	8.23
Aggregate (all sources in each model's val set)	7.763	5.072

The staged schedule not only recovers the flat-mix regression but improves perplexity over the 125M baseline on all three shared domains. We flag one honesty caveat explicitly: the aggregate number looks like a larger win than the per-source numbers alone would suggest, because SEC-EDGAR (inherently more repetitive, formulaic text) makes up roughly 61% of the 500M model's validation set by window count. We treat the per-source numbers, not the aggregate, as the trustworthy signal.

5 Supervised Fine-Tuning

5.1 Data recipe

We combine two complementary sources of supervision into 21,186 instruction pairs (20,127 train / 1,059 held-out validation):

- **Expert-annotated contract data** (ground truth, no synthetic generation): CUAD [Hendrycks et al., 2021b] for clause extraction, and LEDGAR (via LexGLUE [Chalkidis et al., 2021]) for clause classification across 100 categories, both CC BY 4.0.
- **Teacher-distilled question answering**: a locally hosted Meta-Llama-3.1-70B-Instruct model generates grounded question-answer pairs from passages of the pretraining corpus, judged for quality by a combination of NVIDIA Nemotron-Ultra-550B and the same Llama-70B teacher, keeping only pairs scoring $\geq 4/5$.

Source mix: CUAD (~51%), LEDGAR (~22%), SEC filings (~13%), US case law (~12%), educational web (~2.5%). Task mix: extraction (~60%), classification (~22%), question answering (~14%), summarization (~5%), rewriting (<1%).

5.2 A category-name collision failure mode

A handful of clause categories exist under the same name in both CUAD (where the task is "extract and quote the relevant text") and LEDGAR (where the task is "name the category") – for example *Governing Law*, *Effective Date*, *Insurance*, and *Non-Disparagement*. Training on both datasets simultaneously caused the model to collapse to emitting the bare category name for these specific clause types *regardless of which task it was actually asked to perform* – confirmed with a

100% reproduction rate on held-out contracts and hand-written test excerpts before the fix. We resolved this by removing the roughly 400 conflicting LEDGAR examples for the affected categories and reformatting all LEDGAR answers into a structurally distinct template (`"This clause is categorized as: X"`) rather than a bare label, so extraction and classification outputs remain distinguishable throughout training regardless of category-name overlap. We report this in detail because it is a failure mode that would not be visible from aggregate training loss and only surfaced through targeted qualitative inspection of held-out generations.

5.3 Training procedure

Both models are fully fine-tuned (not LoRA) from their respective base checkpoints, at learning rate 3×10^{-5} with cosine decay and 3% warmup, bf16 compute with fp32 master weights, loss masked to answer tokens only. The 500M model's fine-tune ran for 2 epochs; validation loss ticked up between epoch 1 and epoch 2 ($0.1840 \rightarrow 0.1887$), a directly measured overfitting signal on a dataset of this size that we take as evidence against training further epochs without additional data, rather than as a defect to engineer around.

6 In-House Evaluation

Table 4: In-house held-out evaluation (1,059 validation examples), same methodology for both models: extraction/classification scored against ground truth, open-ended question answering judged by an independent local Meta-Llama-3.1-70B-Instruct for semantic correctness.

Task	**125M**	**500M**
CUAD clause extraction (Token-F1)	0.711	**0.761**
CUAD "clause not present" refusal accuracy	**90.1%**	87.7%
LEDGAR clause classification (exact-match)	74.1%	**77.4%**
Case law general QA (closed-book, judged)	35.6%	**42.2%**
SEC filings general QA (closed-book, judged)	43.1%	**46.9%**
Educational web general QA (closed-book, judged)	18.2%	**30.3%**

6.1 Paired comparison: scaling helps on aggregate, not on every question

Aggregate metrics alone can obscure whether a larger model is *strictly* better or merely better *on net*. We additionally ran a paired comparison: both models answering the identical 1,059 held-out questions, with both models' answers freshly re-judged in the same pass to avoid comparing across different judging runs. On CUAD extraction, the 500M model fixed 46 questions the 125M model got wrong, but also newly got 25 questions wrong that the 125M model had answered correctly — a net win, not a clean sweep. One concrete regression, worth citing directly: asked what changed conditions a defendant cited (extractable directly from the given source text), the 125M model correctly listed the real conditions; the 500M model produced a vaguer, legal-sounding but non-responsive answer. Bigger and better on net does not imply strictly better on every question, a distinction we believe matters more for a deployed assistant than the aggregate number alone.

7 External Benchmark Validation

The evaluations above are entirely in-house: our own held-out split, our own extraction/classification scoring, our own judge model. To check the same claims against outside measurement, we additionally evaluate both models with `lm-evaluation-harness` [Gao et al., 2024], the framework underlying a widely used public LLM leaderboard, across five categories, on the deployed fine-tuned checkpoints.

7.1 Task selection

HellaSwag, ARC-Easy, PIQA, and MMLU's `professional_law`, `jurisprudence`, and `international_law` subjects are built into `lm-evaluation-harness` and run unmodified. CaseHOLD and LegalBench required custom task configuration, since neither has an existing harness integration; both configs, along with all raw logs, are released alongside this paper.

For CaseHOLD, we use its LexGLUE packaging (`coastalcph/lex_glue`, config `case_hold`): a 5-way multiple-choice task where the model must identify the correct masked holding statement for a case excerpt. We deliberately excluded LexGLUE's other tasks (SCOTUS, ECtHR, EUR-LEX, UNFAIR-ToS): their documents average tens of thousands of characters (SCOTUS alone averages roughly 70,700 characters, i.e. well over 17,000 tokens in a sampled check), far exceeding this model's 1,024-token context. Truncating a document to under 6% of its length to fit context and then scoring on that truncated prefix would not produce a number comparable to any published baseline. CaseHOLD's excerpts, by contrast, average approximately 850 characters (roughly 200 tokens) and fit comfortably.

7.2 A curated LegalBench subset

LegalBench's 162 tasks span formats (free-text generation, multi-label classification, multiple choice) that do not uniformly fit `lm-evaluation-harness`'s simple multiple-choice scoring, and running the full suite was out of scope for this evaluation pass. We selected 12 tasks matching two criteria: binary yes/no answer format (scoreable as simple 2-way multiple choice) and contract-focused subject matter, matching this project's own CUAD/LEDGAR training domain: nine ContractNLI clause-entailment tasks, `contract_qa`, `consumer_contracts_qa`, and `citation_prediction_classification`. Each task's prompt is the benchmark's own official prompt template (including its baked-in few-shot exemplars), taken verbatim from the LegalBench source repository, not rephrased, so scores reflect the benchmark's intended evaluation format.

7.3 Results

Table 5: General commonsense reasoning (zero-shot, acc_norm). Not a legal claim – a peer-comparison reference point against other small models.

Task	**125M**	**500M**
HellaSwag	29.4%	**32.4%**
ARC-Easy	41.7%	**44.5%**
PIQA	**60.2%**	59.6%

Table 6: MMLU legal subsets (zero-shot, acc). Random-chance floor for 4-choice is 25%; both models sit at or near it.

Subject	**125M**	**500M**
professional_law	24.6%	24.7%
jurisprudence	**30.6%**	25.9%
international_law	**26.5%**	24.0%

Table 7: CaseHOLD (5-way multiple choice; random-chance floor is 20%). Already excluded from both models' pretraining data via 13-gram decontamination, so no contamination concern applies here.

Metric	**125M**	**500M**
acc	11.9%	12.1%
acc_norm	18.4%	19.9%

The 125M model scores *below* the 20% random-chance floor on CaseHOLD's acc_norm metric. Genuinely selecting the correct legal holding among five plausible-sounding alternatives is a categorically different, harder skill than the open-book extraction and classification these models perform well at; we take this as the bluntest possible external confirmation of a capability boundary this project's own model cards had already stated qualitatively.

Table 8: LegalBench, 12-task contract-focused subset (zero-shot, binary yes/no, 50% chance floor). Official LegalBench prompts, verbatim.

Task	**125M**	**500M**
citation_prediction_classification	**54.6%**	50.0%
consumer_contracts_qa	43.2%	**58.6%**
contract_nli_confidentiality_of_agreement	47.6%	50.0%
contract_nli_explicit_identification	**45.9%**	23.9%
contract_nli_limited_use	**61.5%**	60.6%
contract_nli_no_licensing	59.9%	**70.4%**
contract_nli_notice_on_compelled_disclosure	60.6%	**66.2%**
contract_nli_permissible_copy	28.7%	**37.9%**
contract_nli_return_of_confidential_information	**54.6%**	50.0%
contract_nli_sharing_with_employees	50.0%	**53.5%**
contract_nli_survival_of_obligations	61.2%	**71.3%**
contract_qa	**48.8%**	47.5%
Mean across 12 tasks	51.4%	**53.3%**

The LegalBench subset shows a genuinely mixed picture: the 500M model wins 6 of 12 tasks, the 125M model wins 4, and 2 are effectively tied, consistent with the paired-comparison finding in Section 6.1 that scale helps on net without being uniformly better. One result merits a caveat before being read as a real capability gap: on `contract_nli_explicit_identification`, the 500M model scores 23.9%, below its own 50% chance floor, while the 125M model scores 45.9%. With $n = 109$ examples on a single task, we consider this plausibly a prompt-format sensitivity rather than a demonstrated capability regression, and flag it for follow-up rather than drawing a conclusion from it in isolation.

8 A Contamination Case Study: Legal Text Needs a Stricter Check

Unlike CaseHOLD and LexGLUE, which were excluded from this project's pretraining corpus by design, LegalBench was never checked for overlap with the pretraining data before this evaluation. We report the full investigation because we believe the methodological lesson generalizes beyond this specific project: naive n-gram contamination checks are too sensitive for legal text specifically, and the standard fix (a longer n) is necessary but not sufficient without also attributing each surviving hit back to its source.

8.1 A naive check produces an implausible result

Following the same method and threshold this project already uses for CaseHOLD/LexGLUE decontamination (13-word n-gram overlap, 64-core parallel scan across the full ~78GB cleaned pretraining corpus), we built a contamination set from all 12 selected LegalBench tasks' train and test splits (1,853 documents, 120,604 unique 13-grams) and scanned every source. The result was 8,712 overlapping lines across 75 of 76 shard files, spanning every one of the five pretraining sources – including plain educational web text, which has no obvious reason to overlap with contract clause data at all. This was too widespread to trust at face value.

8.2 Inspection: generic legal boilerplate, not leaked test data

Rather than accept or reject the result on its aggregate count, we located and read the actual matched 13-word spans. Every sampled hit was generic, widely recurring legal language: standard-of-review boilerplate that essentially every US appellate court recites verbatim in summary judgment appeals (*"reviewing a motion for summary judgment we apply the same standard as the [trial court]"*); DMCA/Terms-of-Service disclaimer boilerplate reused across thousands of unrelated websites; and canonical precedent-citation language, such as the *Harris v. United States* plain-view-doctrine holding, which any case discussing that doctrine quotes verbatim. None of these are evidence that a specific LegalBench document leaked into training – they are evidence that legal writing recycles fixed phrases far more than general text does, which a blunt phrase-overlap check is not equipped to distinguish from genuine duplication.

8.3 A stricter threshold, plus provenance attribution

We re-ran the check requiring a 40-consecutive-word contiguous match – long enough that coincidental boilerplate reuse cannot plausibly produce it. This reduced the result to 356 hits, concentrated in a small number of files rather than spread across nearly all of them. We then attributed every surviving hit back to its specific source LegalBench document, rather than treating the count alone as a verdict:

In total, 5 of 1,853 evaluated documents (0.27%) show genuine long-run overlap, concentrated in 3 of the 12 tasks. Each is independently explainable by the nature of its source: a famous, widely-republished company Terms of Service document; a standard legal template reused across many real,

Table 9: Attribution of the 356 strict (40-word) contamination hits.

LegalBench task	Docs affected	What it actually is
`consumer_contracts_qa`	3 / 396	A real, publicly available consumer-facing company
`contract_nli_limited_use`	1 / 208	A standard, widely reused NDA template
`citation_prediction_classification`	1 / 108	A real historic court opinion, present in both datasets independently

unrelated contracts; and a historic court opinion that both LegalBench (whose `citation_prediction_classificat` task literally consists of real historical case-law excerpts) and our pretraining corpus (built from the same public case-law record) independently and unavoidably contain. None of this constitutes narrow test-set leakage in the sense that matters for benchmark validity, and none of it is large enough, at under 1% of any single task's examples, to move the reported accuracy outside that task's own margin of error. We consider the LegalBench results in Table 8 safe to report on this basis.

We surface this as a general methodological point: legal text's much higher natural rate of verbatim phrase reuse – case law quoting precedent, contracts reusing boilerplate clauses, companies republishing near-identical Terms of Service – means a contamination-checking threshold tuned for general text will systematically over-flag legal corpora. A longer contiguous-match requirement combined with per-hit source attribution, rather than raw overlap counts alone, is necessary to get a trustworthy signal in this domain.

9 Discussion

Across both the in-house and external evaluations, a single organizing finding recurs: these models are a real, usable capability when given a source passage to work from, and are not a reliable source of legal knowledge or reasoning from parametric memory alone. This is consistent with, and adds an external benchmark confirmation to, the capacity-ceiling argument in Allen-Zhu and Li [2024] – a 528.5M-parameter model can store at most on the order of 132MB of compressible knowledge, shared across grammar, style, and every fact it has ever seen – and the repetition-dependent memorization curve in Carlini et al. [2022], which predicts that facts appearing only once or twice in a multi-billion- token pretraining corpus will be recalled correctly only a small fraction of the time, exactly the regime most specific legal facts fall into.

We deliberately scope this paper to purely parametric capability, without retrieval augmentation [Lewis et al., 2020]. This is not a claim that retrieval would not help – the open-book/closed-book gap documented throughout this paper is, if anything, a direct argument *for* retrieval as a natural next step, since it would convert every closed-book question into the open-book format these models already handle well. We leave that extension, and a systematic comparison against it, to future work, consistent with the recent, larger-scale exploration of retrieval-augmented small legal models in Vaddi [2026].

10 Limitations

- Both models are evaluated against a single held-out validation split per model; we do not report cross-validation or multi-seed variance, so small differences (a few percentage points) between the two model sizes on individual tasks should not be over-interpreted.
- The LegalBench subset covers 12 of the benchmark's 162 tasks, selected for scoring-format and domain fit rather than comprehensiveness; results should not be read as a full LegalBench evaluation.
- SCOTUS, ECtHR, and EUR-LEX (also part of LexGLUE) were not evaluated at all, due to the context-length mismatch described in Section 7, not because they are uninteresting.
- Our contamination check is n-gram-based; it would not detect paraphrased or semantically equivalent (but lexically different) leakage. We consider this an acceptable limitation given the combination of a strict contiguous-match threshold and manual per-hit provenance review, but note it as a real methodological gap.
- Both pretraining runs are single training runs at their respective configurations; we do not report variance across independent reruns of the same recipe.
- Both models are trained on English-language, US-centric legal and financial text; none of the findings here should be assumed to transfer to other jurisdictions or languages.

11 Conclusion

We built two from-scratch small language models for legal and financial text, found and fixed a real data-mixing regression and a real fine-tuning failure mode along the way, and then subjected the resulting models to external, accepted benchmark evaluation rather than relying solely on our own test set. The result is a consistent, externally validated picture: a genuine, narrow capability at open-book contract analysis, and an equally genuine absence of reliable closed-book legal knowledge or reasoning at this parameter count. We release both models, the fine-tuning recipe, an in-browser deployment, and all evaluation code so that these specific, quantified claims – including the ones that do not flatter the larger model – can be independently checked rather than taken on faith.

Reproducibility

Model weights, the fine-tuning dataset recipe, ONNX exports for in-browser inference, and the full pretraining/evaluation codebase (including every custom benchmark task configuration and contamination- check script described in this paper) are available at `https://github.com/DeependraVerma/legal-slm-125M`. Model cards with full evaluation tables are hosted at `https://huggingface.co/DeependraVerma`.

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